Divvela Rakesh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.68
Narayana Junior College	Class XII (TSBIE)	2023	98.00%
Narayana High School	Class X (SSC)	2021	10.00%

WORK EXPERIENCE

Summer Analyst, Goldman Sachs | **2026**

- Refactored document processing into a **configuration-driven framework**, enabling operations such as citation generation and validation to be configured via YAML and simplifying onboarding of new document types.
- Designed a **modular LLM generation pipeline** for multi-document financial analysis, supporting configurable field mapping, calculations, and narrative generation.
- Implemented **parallel AWS Lambda-based document processing**, replacing sequential document merging and reducing latency for multi-document analysis.

Teaching Assistant for Database Management Systems and Computer Networks

PROJECTS

Delay-Compensated Interval Asynchronous SGD *(Research Work /Prof. Satya Peri)*

- Implemented a **lock-free parallel SGD** framework in **C++** using **OpenMP**, with thread-local mini-batch computation and atomic updates to shared model parameters.
- Integrated **delay compensation** with **interval-based asynchrony**, bounding gradient staleness and correcting stale updates to reduce optimization drift during lock-free training.
- Reduced final training loss by **~43%** over Interval-ASGD at **64** threads through delay-compensated asynchronous training.
- Adapted an automatic interval-probing mechanism for high-contention workloads, achieving a final loss of **~0.25** at **128** threads, compared with **~1.57** for **DC-ASGD**.

TLS Misconfiguration Scanner *(Course Project/Prof .Kotaro/Prof Praveen)*

- Developed a modular Python-based TLS scanner with **11** implemented check classes covering **24** logical checks for protocol versions, cipher suites, certificates, session tickets, ALPN, OCSP, delegated credentials, and TLS vulnerabilities.
- Implemented **behavioral and adversarial TLS checks** including downgrade resistance, insecure renegotiation, STEK misuse, TicketBleed, and Delegated Credential validation using OpenSSL-based handshake probes.
- Built multiprocessing-based parallel scanning, reducing runtime from **~130s** to **~4s** for **20** targets (**~32 x** faster) and scaling scans to **100** targets in **~36.5s**.

Multi-Client Quiz System with LLM Integration *(Course Project /Prof .Kotaro/Prof Praveen)*

- Developed a **TCP-based multi-client** quiz system in **C++** with LLM integration for real-time question generation, answer evaluation, and leaderboard management.
- Designed a **custom application-layer protocol** with typed message framing and keep-alive messaging for **session management**.
- Implemented server-side **FIFO caching** to reduce redundant LLM requests for frequently requested quiz genres, and analyzed throughput and request-response latency under varying client concurrency using Wireshark

Virtual Memory Management System in gemOS *(Course Project / Prof.Saurabh kumar)*

- Implemented memory management system calls (**mmap, munmap, mprotect**) with dynamic virtual memory management in C.
- Designed **lazy allocation** mechanism with page fault handling, achieving efficient memory usage through **on-demand** physical page allocation.
- Built **4-level** page table architecture with **Page Table Entry** (PTE) management and virtual-to-physical address translation.

RISC-V Assembly Simulator with Cache *(Course Project/ Prof. Rajesh)*

- Developed an **RV64I simulator** in C++ supporting step-by-step execution, register state tracking, and program control.
- Implemented **breakpoint-based debugging**, enabling execution to pause and resume at user-defined breakpoints.
- Built a **configurable** cache simulator for memory operations with cache statistics, supporting LRU, LFU, and Random replacement policies across varying cache sizes and associativities.

SKILLS

Programming Languages: C/C++,Python,JavaScript

Tools & Technologies : MySQL,Git,Docker,GitLab,Bash,Confluence,Latex

RELEVANT COURSES

CSE Core Courses: Data Structures, Operating Systems, Database Management, Computer Architecture, Computer Networks, Compilers, Foundations of Machine Learning, Deep Learning, Quantum Computing

Mathematics: Linear Algebra, Probability and Random Variables, Differential Equations, Group Theory

SCHOLASTIC ACHIEVEMENTS

- **NSEP** : Certificate of Merit - Top **1%** of 41,454 candidates (National Topper)
- **NSEC** : Certificate of Merit - Top **1%** of 36,831 candidates (National Topper)

POSITIONS OF RESPONSIBILITY

- Sci-Tech core at Blockspace
- UG Mentor at Sunshine Counselling Cell

EXTRACURRICULAR

National Service Scheme (NSS) -Student Volunteer